

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming packets that do not belong to an established, active connection. Which firewall type is best suited for this task? (1 Marks)

- A. Stateless Packet Filter
- B. Stateful Inspection Firewall
- C. Application-Level Gateway
- D. Circuit-Level Gateway

Q2. An organization wants to inspect the actual payload of HTTP traffic to prevent SQL injection attacks. Which firewall type is best suited? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-Level Gateway
- C. Application-Level Gateway
- D. Stateful Inspection Firewall

Q3. A firewall operates at the Session layer of the OSI model, verifying the TCP handshake without inspecting the packet payloads. What type of firewall is this? (1 Marks)

- A. Packet Filter
- B. Circuit-Level Gateway
- C. Application Proxy
- D. Next-Generation Firewall

Q4. Which feature primarily distinguishes a Next-Generation Firewall (NGFW) from a traditional stateful firewall? (1 Marks)

- A. Port-based blocking
- B. Deep Packet Inspection (DPI) with application awareness
- C. TCP handshake verification
- D. IP address filtering

Q5. A company needs to protect its web servers specifically from Cross-Site Scripting (XSS) and SQL injection. Which specialized firewall should they deploy? (1 Marks)

- A. Network-level Packet Filter
- B. Web Application Firewall (WAF)
- C. Circuit-Level Gateway
- D. Stateful Inspection Firewall

Q6. Why is a stateless packet filtering firewall considered vulnerable to IP spoofing attacks? (1 Marks)

- A. It does not inspect the IP header.
- B. It evaluates packets individually without context of the connection state.
- C. It decrypts SSL traffic automatically.
- D. It operates only at the Application layer.

Q7. An administrator wants to protect individual endpoints from lateral movement within the internal network. Which deployment is most appropriate? (1 Marks)

- A. Perimeter Network Firewall
- B. Host-based Firewall
- C. Circuit-Level Gateway
- D. Bastion Host

Q8. Where should a public-facing web server be placed to minimize risk to the internal corporate network? (1 Marks)

- A. Inside the internal intranet
- B. Directly on the public Internet
- C. Within a Demilitarized Zone (DMZ)
- D. Behind a stateless packet filter only

Q9. When a firewall terminates an IPsec VPN tunnel, at which layer of the OSI model is the encryption/decryption primarily handled? (1 Marks)

- A. Physical Layer
- B. Network Layer
- C. Session Layer
- D. Application Layer

Q10. How does a firewall utilizing Port Address Translation (PAT) conserve public IP addresses? (1 Marks)

- A. By assigning a unique public IP to every internal host
- B. By mapping multiple private IP addresses to a single public IP using unique source ports
- C. By blocking all outbound UDP traffic
- D. By routing traffic strictly through an application proxy

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Stateful inspection firewalls track the state of active connections to determine whether to allow incoming packets.

Q2. Correct Answer: C

Rationale: Application-level gateways inspect the application layer payload to detect deep content threats.

Q3. Correct Answer: B

Rationale: Circuit-level gateways operate at the session layer to monitor TCP handshaking and validate sessions.

Q4. Correct Answer: B

Rationale: NGFWs integrate deep packet inspection and application-level control beyond traditional port and protocol filtering.

Q5. Correct Answer: B

Rationale: A WAF is specifically designed to analyze HTTP/HTTPS traffic at the application layer to protect web applications.

Q6. Correct Answer: B

Rationale: Stateless firewalls inspect packets in isolation, making them unable to verify if a packet is part of an ongoing legitimate session.

Q7. Correct Answer: B

Rationale: Host-based firewalls run locally on individual devices to secure them against internal network threats.

Q8. Correct Answer: C

Rationale: A DMZ isolates public-facing services from the sensitive internal network.

Q9. Correct Answer: B

Rationale: IPsec operates at the Network Layer (Layer 3) of the OSI model to secure IP communications.

Q10. Correct Answer: B

Rationale: PAT maps multiple internal private IP addresses to a single public IP address by tracking unique source port numbers.